

Lab 05

Due March 7, 5:00 PM PST

AUTHOR

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Modify this template file to create your submission.

```
## write answer here
library(tidyverse)

# Load the bird sightings data and species names
ebird <- read.csv("ebird30.csv", stringsAsFactors = FALSE)
names_df <- read.csv("names.csv", stringsAsFactors = FALSE)
```

Problem 1a

```
regions <- c("CA.MB", "CA.BC", "CA.AB", "CA.SK", "CA.ON",
            "CA.QC", "CA.NL", "CA.NS", "CA.NB", "CA.PE",
            "CA.NT", "CA.NU", "CA.YT")

region_counts <- setNames(rep(0, length(regions)), regions)

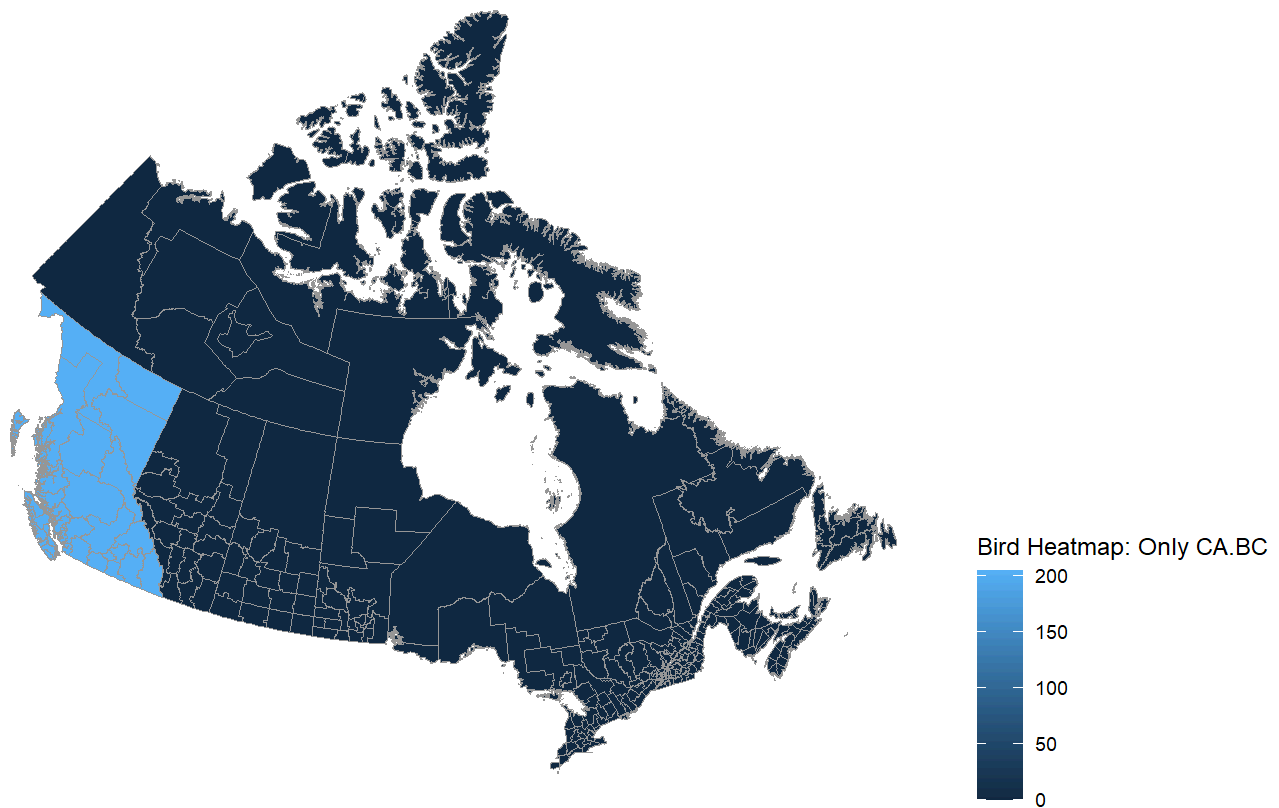
# Loop through each species (row in ebird) and count nonzero sightings per region
for (i in 1:nrow(ebird)) {
  for (region in regions) {
    count <- as.numeric(ebird[i, region])
    if (!is.na(count) && count > 0) {
      region_counts[region] <- region_counts[region] + 1
    }
  }
}

source('canada.R')

# Create a heat list with zeros for all regions
heat <- as.list(setNames(rep(0, length(regions)), regions))

# Set only CA.BC to its count (or any nonzero value you wish to highlight)
heat[["CA.BC"]] <- region_counts[["CA.BC"]]

title <- "Bird Heatmap: Only CA.BC"
canada(title, heat)
```



Problem 1b

write answer here

```
ebird$total <- rowSums(ebird[, -1], na.rm = TRUE)
```

```
max_species <- ebird[which.max(ebird$total), ]  
max_species <- max_species$Code
```

```
res <- names_df[names_df$Code == max_species, ]  
res$Code
```

```
[1] "cangoo"
```

```
res$Common.Name
```

```
[1] "Canada Goose"
```

```
res$Scientific.Name
```

```
[1] "Branta canadensis"
```

Problem 1c

```
## write answer here
smallest_count <- ebird[ebird$total != 0, ]
smallest_count_num <- min(smallest_count$total)
smallest_count <- smallest_count[smallest_count$total == smallest_count_num, ]

smallest_count$common_name <- names_df$Common.Name[match(smallest_count$Code, names_df$Code)]

sorted_birds <- sort(smallest_count$common_name)
head(sorted_birds, 10)
```

```
[1] "American Wigeon x Mallard (hybrid)" "Blue-winged Teal"
[3] "Broad-winged Hawk"                "Clay-colored Sparrow"
[5] "Common Yellowthroat"              "Dickcissel"
[7] "Dusky Grouse"                     "Golden Pheasant"
[9] "Graylag x Swan Goose (hybrid)"     "Great Egret"
```